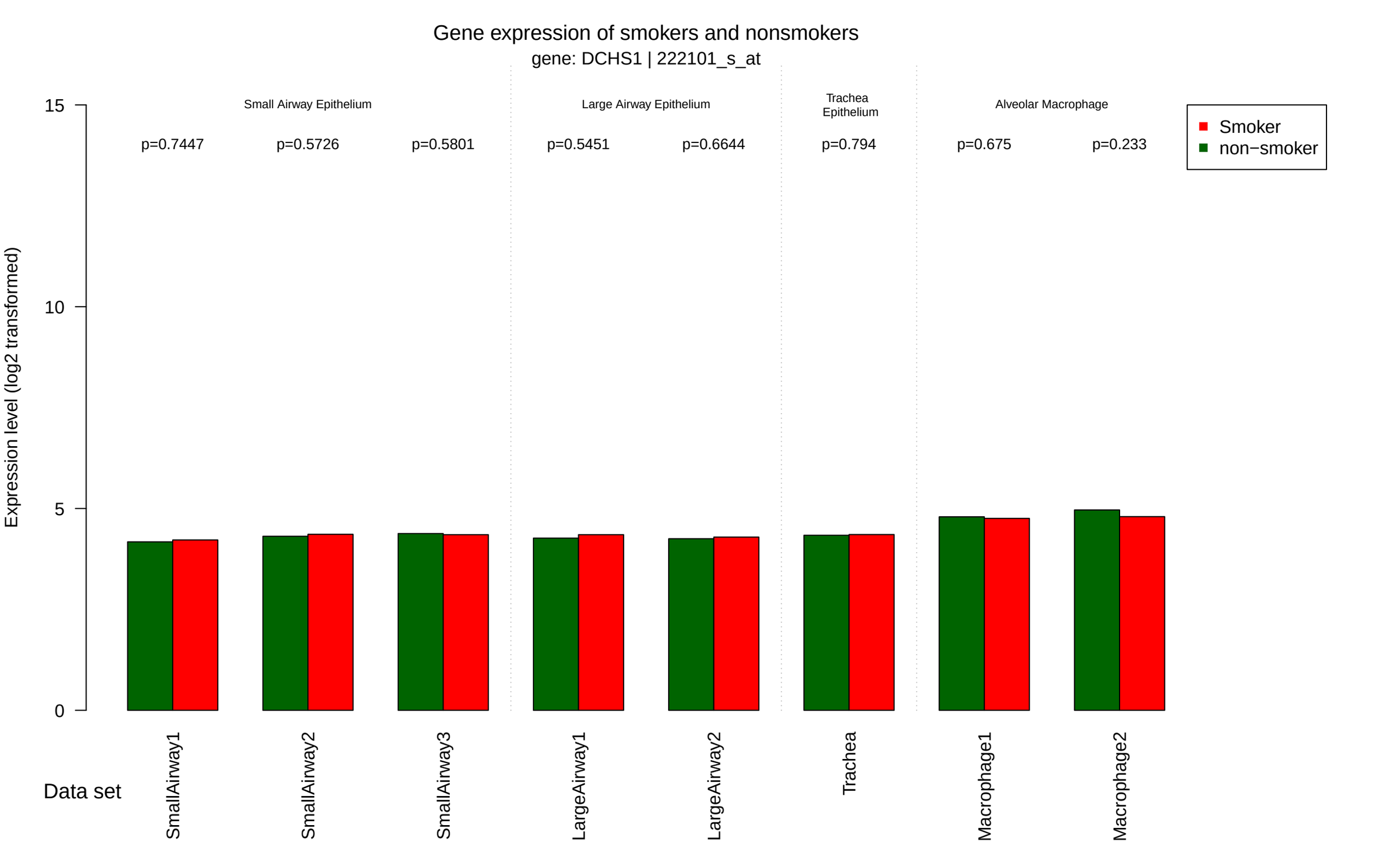
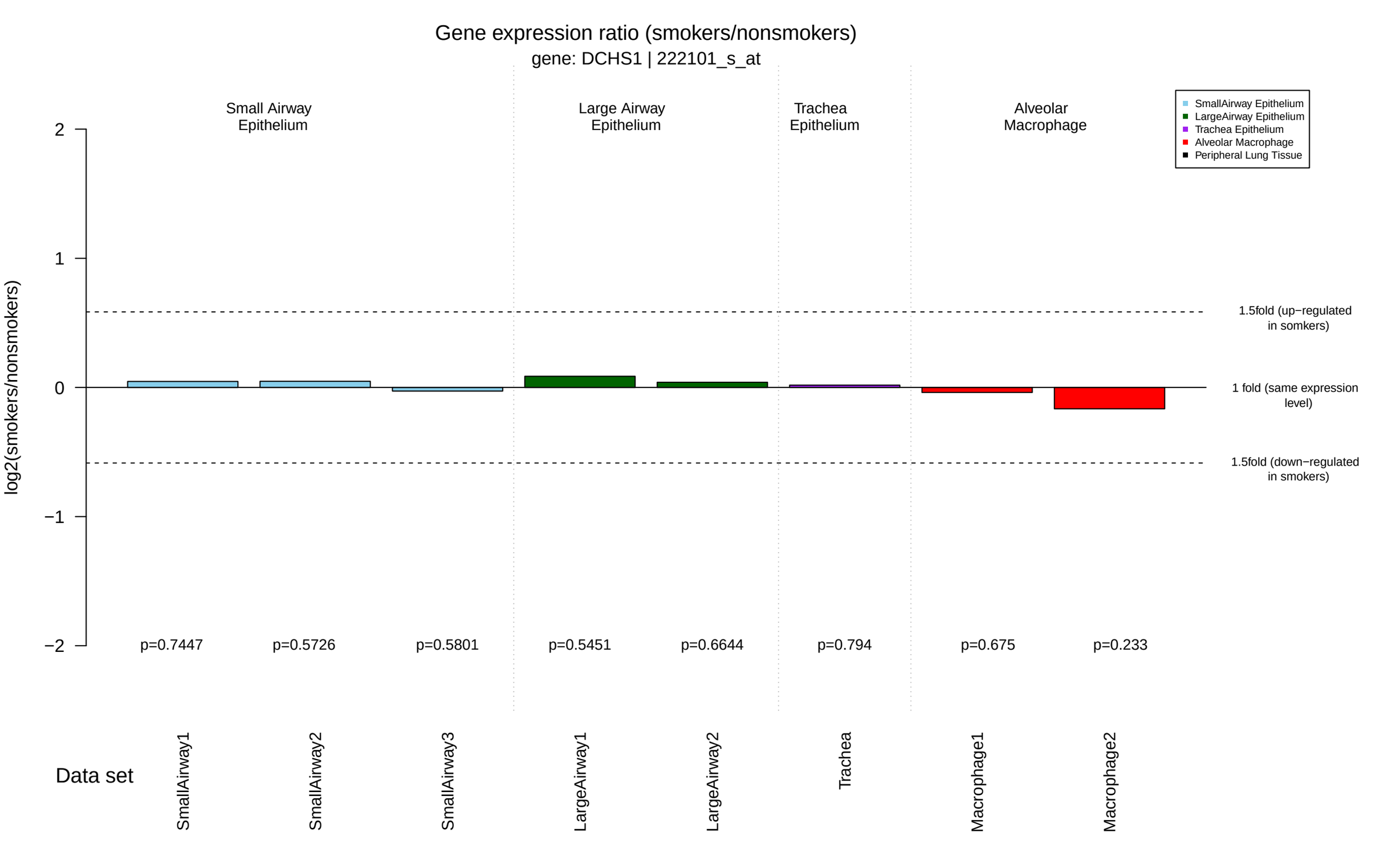
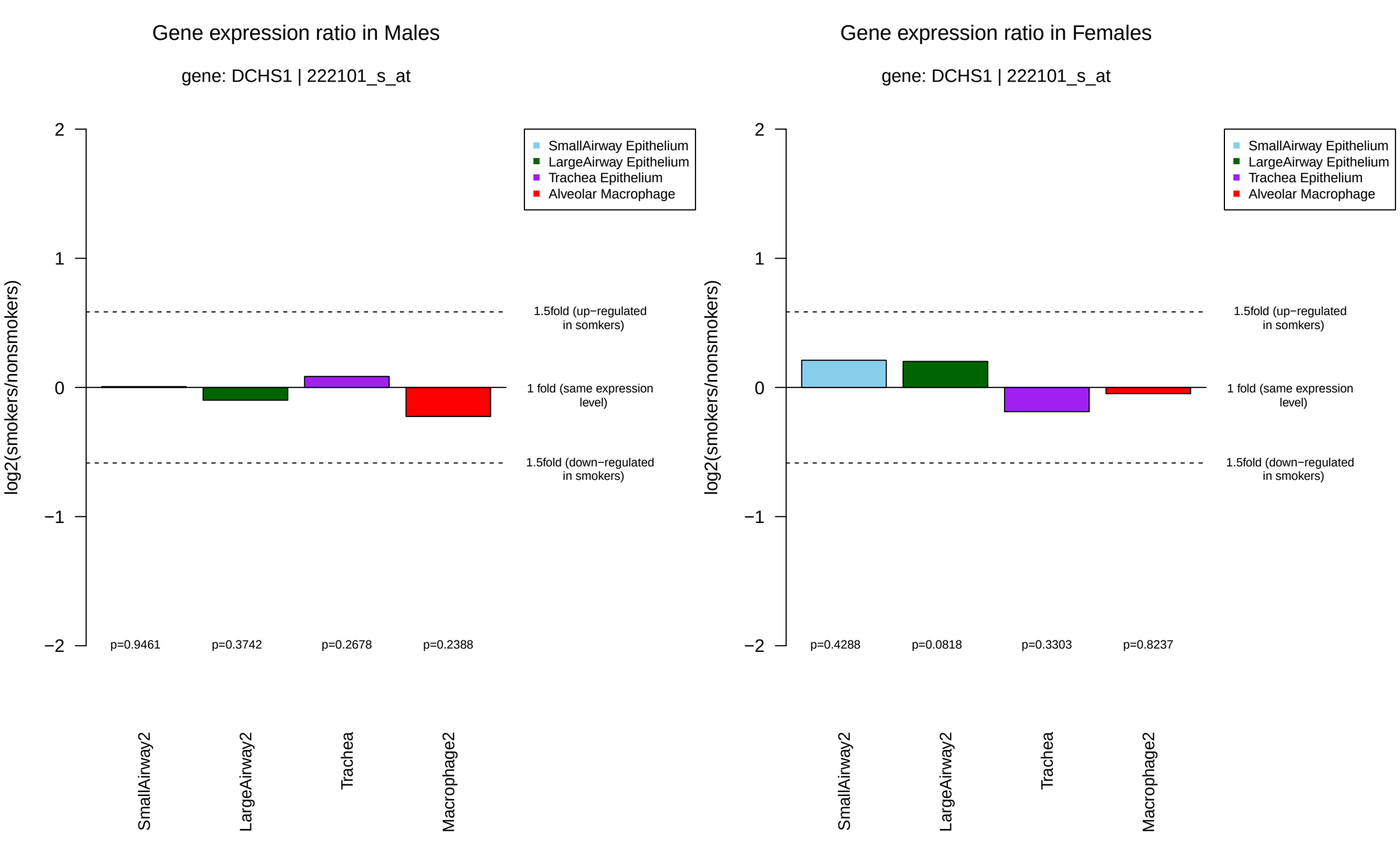
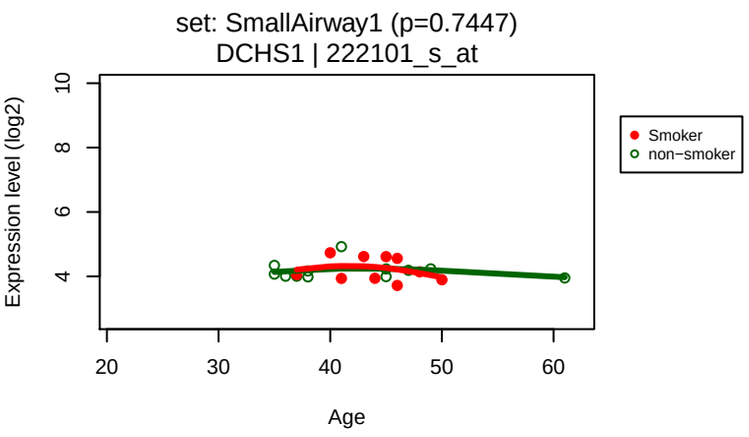
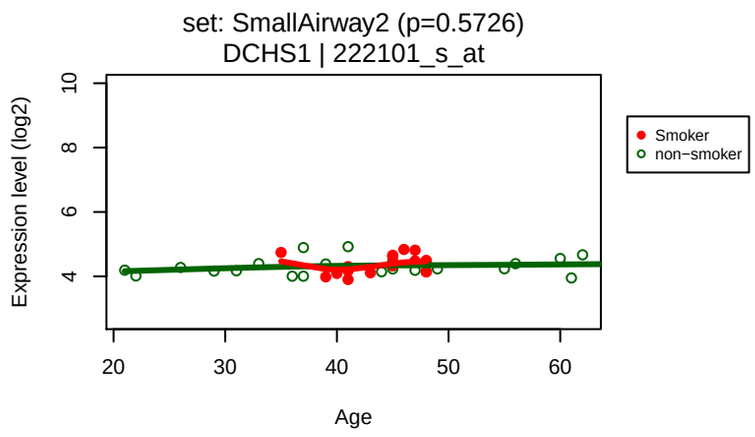
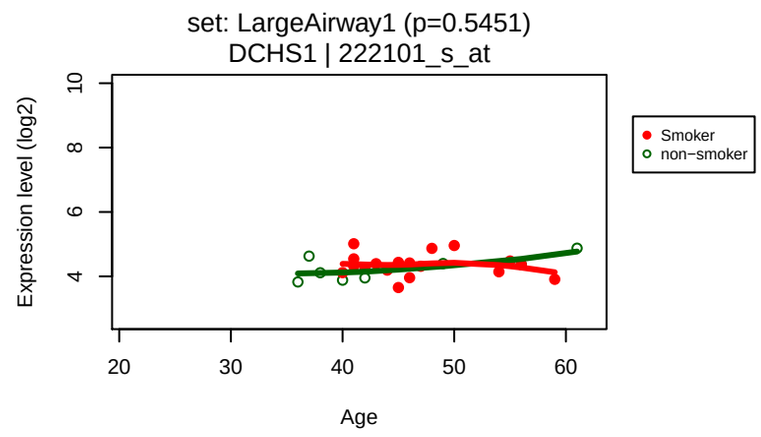

Gene expression of smokers and nonsmokers
gene: DCHS1 | 222101_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7447
p=0.5726
p=0.5801
p=0.5451
p=0.6644
p=0.794
p=0.675
p=0.233
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

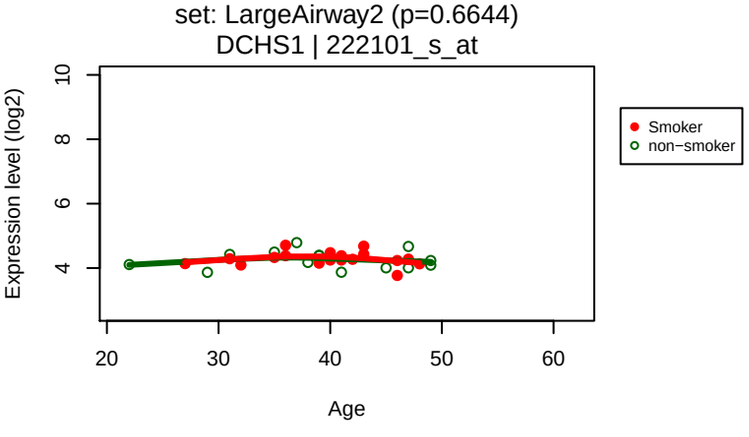

Gene expression ratio (smokers/nonsmokers)
gene: DCHS1 | 222101_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7447
p=0.5726
p=0.5801
p=0.5451
p=0.6644
p=0.794
p=0.675
p=0.233
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

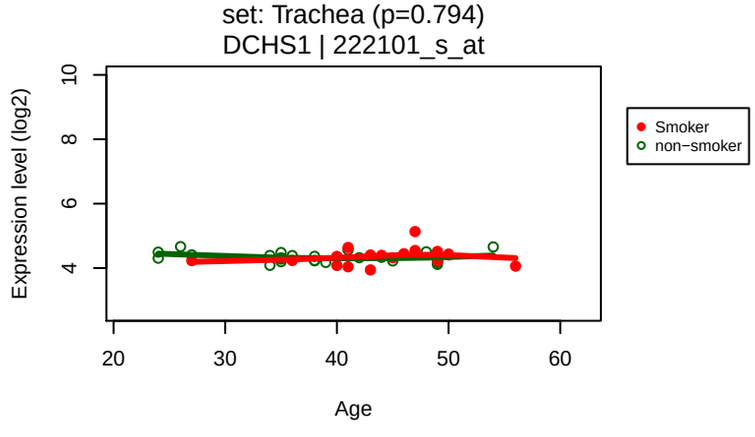

Gene expression ratio in Males
gene: DCHS1 | 222101_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9461
p=0.3742
p=0.2678
p=0.2388
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DCHS1 | 222101_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4288
p=0.0818
p=0.3303
p=0.8237
SmallAirway2
LargeAirway2
Trachea
Macrophage2

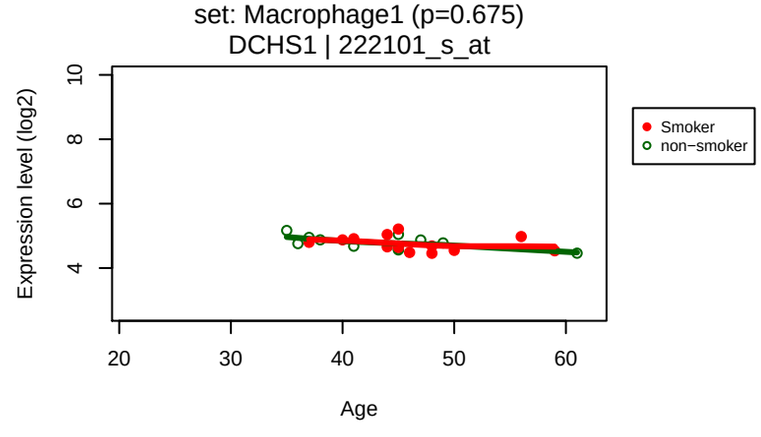

set: SmallAirway1 (p=0.7447)
DCHS1 | 222101_s_at
Expression level (log2)
Age
Smoker
non−smoker

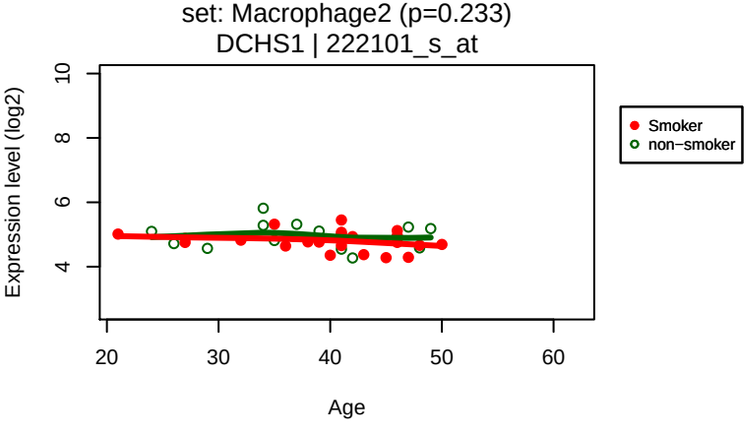

set: SmallAirway2 (p=0.5726)
DCHS1 | 222101_s_at
Expression level (log2)
Age
Smoker
non−smoker

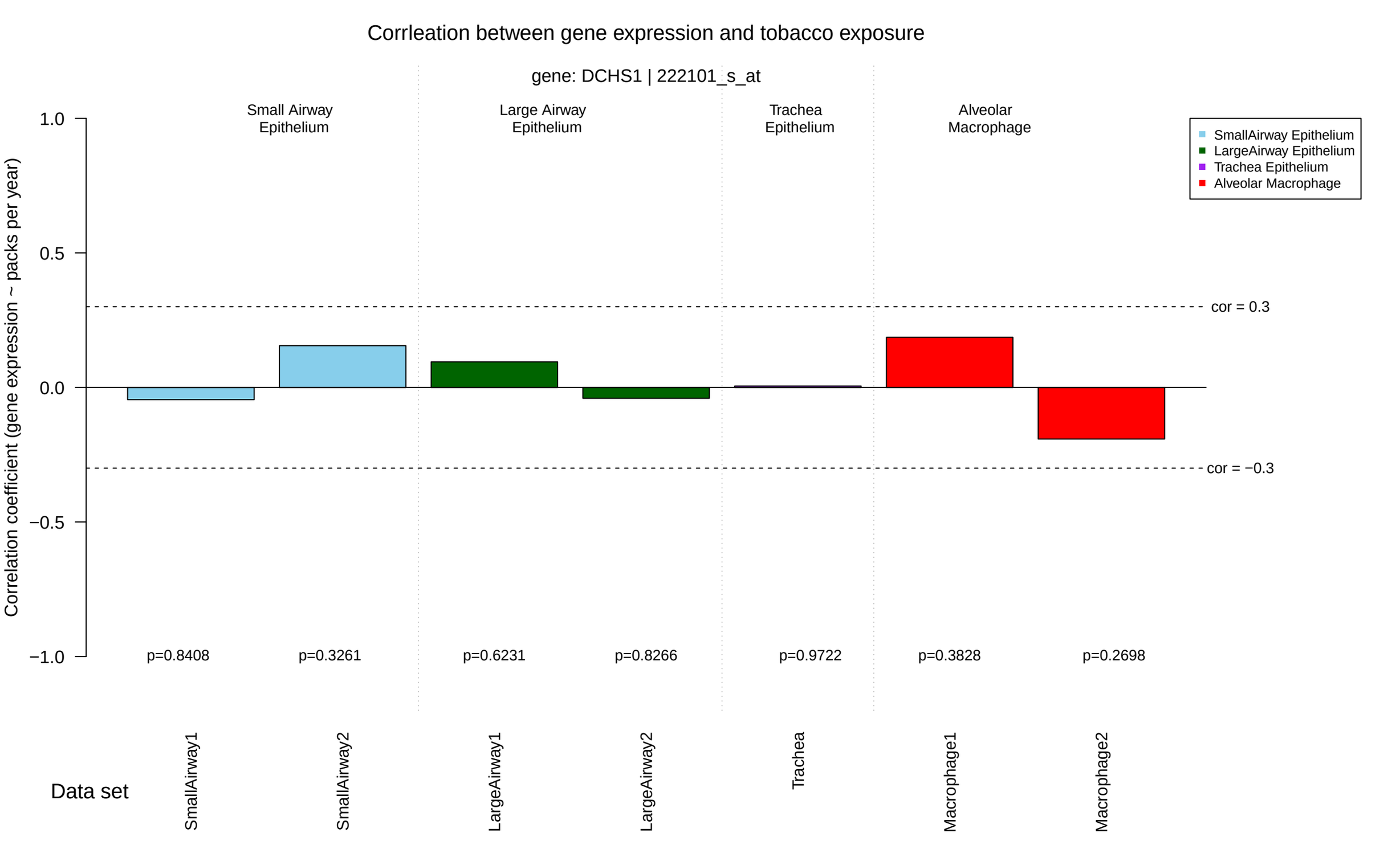

set: LargeAirway1 (p=0.5451)
DCHS1 | 222101_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6644)
DCHS1 | 222101_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.794)
DCHS1 | 222101_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.675)
DCHS1 | 222101_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.233)
DCHS1 | 222101_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DCHS1 | 222101_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8408
p=0.3261
p=0.6231
p=0.8266
p=0.9722
p=0.3828
p=0.2698
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2